

Laseroma: A Small-Sized, Light-Weight, and Low-Cost Olfactory Display Releasing Multiple Odors through Pointed Heating

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ABSTRACT

In this paper, we propose *Laseroma*, a tiny, lightweight, and low-cost Olfactory Display (OD) that can release a maximum of eleven types of odors through pointed heating. *Laseroma* can be easily attached to Head-Mounted Devices (HMDs) for creating smell-enhanced and controllable Immersive Experiences. *Laseroma* mainly consists of (1) a laser diode for rapidly generating the pointed heat, (2) a modular and replaceable odor strip which carries eleven miniature odor reservoirs, and (3) a stepper motor for rotating the odor strip for switching the target aroma. We illustrated and articulated the system design, including the hardware, structural mechanism, materials, and fabrication process. We also conducted a preliminary study by testing the scent release performance through the perceived time and odor-release duration of a single scent container.

CCS CONCEPTS

• Human-centered computing → Interaction devices.

KEYWORDS

olfactory display; odor strip; pointed laser heating; odor switching; controllable scent releasing; virtual reality

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1 INTRODUCTION

As one of the essential sensations among the five senses, the human sense of smell plays a crucial role in influencing human behavior[4], emotions[10], and cognition[5]. In Human-Computer Interaction (HCI), scent-releasing technology has been increasingly favored, particularly in creating complex and immersive smell experiences, such as gaming[8], training[11], in-car interactions[2], and odor reproduction[7]. However, scent-delivery devices face limitations

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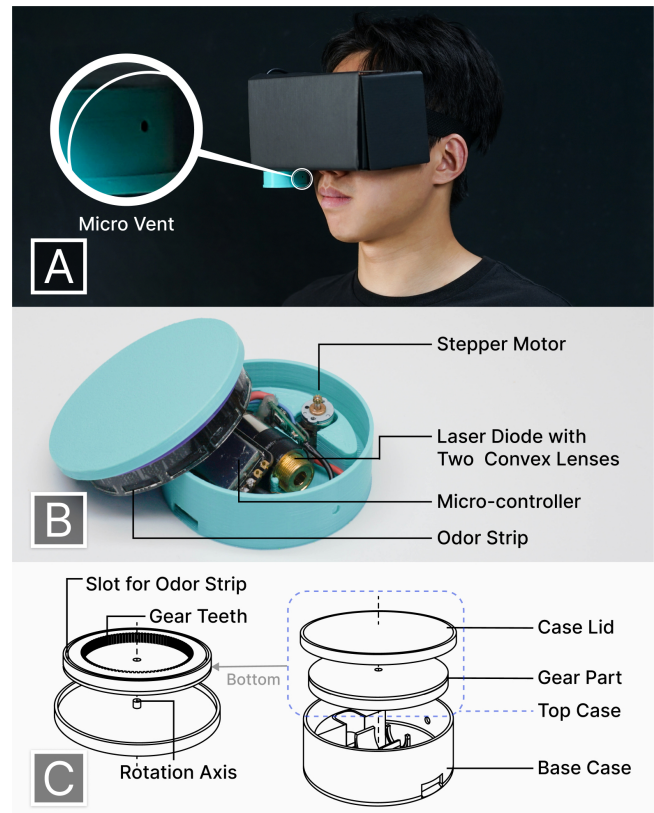


Figure 1: *Image A: Laseroma* can be easily attached to a VR headset. *Image B* shows the inner structure and electronics of *Laseroma*. *Image C* illustrates the explosion diagram of *Laseroma*. We printed it using an Anycubic 3D printer. The whole weight of *Laseroma* is 33g with a diameter of 52mm and height of 18mm. We control *Laseroma* with a microcontroller (Seed XIAO). All the materials were purchased from online shops, costing approximately 90 CNY (12 dollars).

of dimension and weight due to odor's inherent properties (e.g., the absence of primary aromas). Some designers and researchers have attempted to explore small-sized odor-release devices that contain multiple odors, such as a soft mask-like wearable[6] that contains 9 odors, a neck-worn device with 8 fragrances[3], and a smell tape releasing scheduled scents through scratching[1]. In this paper, we further propose a controllable, tiny, light-weight, and low-cost olfactory display called *Laseroma*, which can release

11 different types of aromas using only one heating module, laser diode (See Figure 1). It can accommodate 11 or even more odors in a compact size and ensure a high usage time (more than 40 times with 5 seconds of heating through evaluation). We hope that *Laseroma* can assist head-mounted displays in providing more user-friendly, sophisticated, and dynamic smell experiences within immersive environments.

2 IMPLEMENTING LASEROMA

Modular and Replaceable Odor Strip. The odor strip comprises a 0.5mm thick PET sheet and a high-temperature-resistant PTFE tape (Polytetrafluoroethylene[9], thickness = 0.14mm). We first cut the PET sheet into ribbons (134mm × 10mm) with 11 rectangular hollows (10mm × 2mm) using a CNC machine and then attached the strip to the PTFE tape, shaping the miniature odor containers. More scents can be achieved with smaller reservoirs. We intentionally chose a black color PTFE to ensure better heat and light absorption. We scratched the wax-based odor source inside the reservoirs and tightly inserted the odor ribbon into the slot of the gear part (Figure 1-C). We face the scent containers toward the outer side to prevent odor from mixing within the inner space (Figure 1-B). We devised the odor strip modular and replaceable for scenario-based customization.

Laser-Heating Module. We constructed a close-range and single-point heating structure utilizing a micro laser diode (wavelength = 625nm, maximum output power = 250mW) and two convex lenses (focal length = 15mm). We set the operating voltage and current of the laser to 3.3V and 0.16A to avoid ignition. We control this module by connecting it to a laptop or smartphone via a USB-C for power supply. The concentration and duration of odor release can be controlled and modified by adjusting the heating time of the laser diode. Under these parameters, the temperature of the micro odor container will rise to 75°C when heating for 5 seconds (measured using a Probe Thermometer). We localized the heating module in the center of the base case for space-saving considerations. Notably, even high temperature is achieved, the wearers will not perceive the thermo because of the inner structure design and the non-skin placement.

Odor Switching. The odor switching relies on a miniature stepper motor with a step angle of 18 degrees (2-phase 4-wire, diameter = 8mm) to rotate the odor strip. The motor drives the internal gear and the odor ribbon to adjust the target odor. We measured that the motor takes less than 1s to rotate within two containers (1.2cm length) under the present gear mechanism, and the maximum time cost is highly relevant to the half length of the odor ribbon by setting clockwise and counterclockwise rotation. After the rotation adjustment, the laser will automatically project the heat, and the scented air will be extruded outside the case through the micro vent (Figure 1-A).

3 EVALUATION

We experimented in an open office environment at a room temperature of 26°C to evaluate the *scent release performance*, *duration of a single odor container*, and the *noise level of the rotating system*. We recruited 8 participants (6 males) from campus with an average age

of 24.38 years (SD = 1.22) and obtained consent forms from each participant. We chose a gardenia-scented source for the odor-release test. We pre-set the heating time of *Laseroma* for 5 seconds.

Experiment Setup and Tasks. *Firstly*, the experimenters positioned *Laseroma* near each participant's cheek and started to heat, approximately 2.5cm away from the nose (Figure 1-A). We repeated this action ten times with a 20-second interval for cooling. The participants were required to rate their smell sensitivity and perceived intensity using a 5-point Likert scale (1 = weakest, 5 = strongest) and informed the experimenters immediately after they smelled the released scents. The experimenters recorded the perceived time from the laser started to heat to the participants' detection of odors with a phone-based timer. *Secondly*, we prepared two different samples for the duration test. Sample A used a brand-new odor strip filled with a gardenia aroma, while Sample B was inserted with the same odor but heated over 40 times. To decrease the bias, we performed a blind test without informing the participants about the samples' details. The participants were required to smell the non-heating samples and report the sample with weaker intensity to the experimenters.

Results. Our evaluation showed that the participants (smell sensitivity: $M = 4$; $SD = 1.19$) could moderately perceive the released smells ($M = 3.0$; $SD = 0.66$) with an average perceived time of 2.68 seconds. For the duration test, by choosing the sample with weaker intensity, only two participants selected Sample B, three chose Sample A, while another three users reported no difference between the two samples. We could not find a noticeable decrease in odor concentration through the test. Additionally, we measured the noise generated by the stepper motor when driving the internal gear using the Decibel X app at a distance of 10cm (approximately equal to the distance from the device placement to the ear). The results showed that the noise level was 38.5 dB with an ambient noise of 34dB, demonstrating that *Laseroma* could create relatively silent experiences even with mechanical motion.

4 FUTURE WORK AND CONCLUSION

Our experiment demonstrated that even with minimal scented source (10mm³), *Laseroma* can achieve a long-term duration of scent release and provide controllable smell experiences with a perceived time of 2.68s. For future work, we will first iterate the gear mechanism and heat-resistant material (stand higher temperature) to achieve high-speed scent switching and a shorter perceived time for real-time feedback. We will also optimize the structural design to prevent odor mixing and leakage before being released and conduct a full user study to evaluate the smell experience in a virtual environment.

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